For A any $m \times n$ real matrix we want to write

$$A = U \Sigma V^T$$

U: $m \times m$ orthogonal
Σ: $m \times n$ "diagonal"
V: $n \times n$ orthogonal

This is in a sense a generalized form of orthogonal diagonalization. The outer two matrices are both orthogonal and middle one is diagonal just as in that.

Here though Σ is not usually square, so certainly not a "traditional" diagonal matrix. Similar but also definitely different as well.

Example last time with classes + times, ranking based combination of class + time, well if you are considering ten classes amongst eight different time

slots, if one of those represents the U matrix, the other the V matrix, well then U and V won't be of the same size, you don't always have control over that. Hence why we might want to expand on ideas of diagonalization to all matrices.

Big thing is that $U, V,$ although orthogonal, are not inverses of each other, in fact again usually they are not even same size. So therefore this isn't as "special" as orthogonal diagonalization.

That's the tradeoff naturally, not as special, but it works for any matrix unlike orthogonal diagonalization which only works for symmetric (therefore square inherently required) matrices.

U and V are in a sense completely disconnected. That is, if we pick U to be an $m \times m$ orthogonal matrix, and then independently pick V to be an $n \times n$ orthogonal matrix while choosing Σ to be any diagonal matrix

of the proper size, whatever that product becomes, this will be a SVD for that matrix even though U and V were chosen completely independently.

Of course when finding SVD for some fixed A, the U and V are inherently related since they need to come together to create A so in that sense they do affect each other. But again even in creating regular diagonalization, PDP^{-1}, once P is picked certainly P^{-1} is totally forced, not like earlier case when you can pick the two outer matrices independently.

Now again how do we know such a breakdown exists and how to find it if so?

Start with $A^T A$ which is an $n \times n$ symmetric matrix that importantly has only real, nonnegative eigenvalues. Symmetric via

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

Only real eigenvalues was shown in HW 5.5 (mentioned this also when we did orthogonal diagonalization).

Only nonnegative eigenvalues is new though. Important to note that that doesn't hold for every symmetric matrix, only for ones of form $A^T A$. After all, any diagonal matrix with at least one negative diagonal entry would be symmetric with at least one negative eigenvalue.

Why is that true? Well say λ is an eigenvalue with eigenvector v. Then observe that

$$\underbrace{(Av)\cdot(Av)}_{\geq 0} = (Av)^T(Av) = v^T A^T \underbrace{Av = v^T \lambda}_{\text{eigenvector use}} v = \lambda v^T v$$
$$= \lambda \underbrace{(v \cdot v)}_{\geq 0}$$

Since both those are ≥ 0, forces $\lambda \geq 0$ as desired.

Thus $A^T A$ has n (not necessarily distinct) real, nonnegative eigenvalues. Order them in decreasing order and say the square roots of those are:

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > \sigma_{r+1} = \ldots = \sigma_n = 0$$

We call $\sigma_1, \sigma_2, \ldots, \sigma_n$ the <u>singular values</u> of A.

We use the square roots instead of the originals because that turns out to be easier notationally. Also it helps emphasize that the eigenvalues are nonnegative.

So the first r singular values are positive, the remaining $n - r$ are 0. We don't know what r equals (depends on A) but as we will see later, the associated eigenvectors for these have to be handled differently depending on whether they correspond to an eigenvalue of 0 vs. a positive eigenvalue so important to make a distinction.

So are the singular values of A the square roots of the eigenvalues of A? No! That's a common misconception. A after all is not necessarily square, it often won't even have eigenvalues and even if it did, no guarantee they are real and nonnegative.

Singular values of A are square roots of eigenvalues of $A^T A$ not A! So every matrix, square or not, has singular values unlike eigenvalues which only square matrices have.

So $A^T A$ has n real, linearly independent eigenvectors (it is symmetric). Make those orthonormal (same idea as orthogonal diagonalization, mostly the eigenvectors will be orthogonal already, only some eigenspace(s) with dimension higher than 1 might require Gram-Schmidt).

Denote that set of orthonormal eigenvectors as $\{v_1, v_2, \ldots, v_n\}$

where each v_i corresponds to eigenvalue σ_i^2

Important properties these have are:

(1) $\|Av_i\| = \sigma_i \quad \forall \; 1 \le i \le n$. Why?

$$\|Av_i\|^2 = (Av_i)\cdot(Av_i) = (Av_i)^T(Av_i)$$

$$= v_i^T(A^TAv_i)$$

$$= v_i^T\sigma_i^2 v_i$$ ----> since eigenvector

$$= \sigma_i^2(v_i^T v_i) = \sigma_i^2$$

since each v_i has length 1.

Thus $\|Av_i\| = \sigma_i$

(2) For $i \neq j$, $(Av_i) \cdot (Av_j) = 0$. Why?
Similar to (1) we have

$$(Av_i) \cdot (Av_j) = \sigma_j^2 (v_i^T v_j) = 0$$

since v_i, v_j are orthogonal when $i \neq j$.

Now for each $\sigma_i \neq 0$, i.e. $1 \leq i \leq r$, take

$$u_i = \frac{1}{\sigma_i} Av_i$$

Hence why we wanted to draw distinction between positive vs. 0 singular values as this of course would not make sense for $\sigma_i = 0$.

By the two prior statements, the set of u_i's are an orthonormal set in $\mathbb{R}^m$ (since length of each Av_i is σ_i and the Av_i's are orthogonal).

But there are only r such u_i's and we really need the full m of them, a basis for $\mathbb{R}^m$. Since we don't what r is (depends on A), this is not guaranteed.

For instance, there are at most n of the u_i's at the start since there are exactly n of the v_i's. So if say $m > n$ we will definitely need to do this since we need m of the u_i's.

Hence we might need to extend this set of u_i's to an orthonormal basis for $\mathbb{R}^m$, denoted

$$\{u_1, u_2, \ldots, u_r, u_{r+1}, \ldots, u_m\}$$

How to do this? Variety of ways to extend basis, then might have to use Gram-Schmidt to turn it orthogonal.

In essence, if we think of $\{u_1, u_2, \ldots, u_r\}$ as generating a subspace W, we want to find an orthonormal basis for $W^\perp$.

So one option to do that is to make the current u_i's the rows of a matrix, and then to find the null space of that matrix say via row echelon form + using parametric vector form, that will generate the $W^{\perp}$ that we need (HW 6.3.32)

But that basis we find might not be orthonormal, might need to use Gram-Schmidt to make it so.

So with that take U to be the $m \times m$ matrix whose columns are: $u_1, u_2, \ldots, u_m$

Then take V to be the $n \times n$ matrix whose columns are $v_1, v_2, \ldots, v_n$.

Note that we can also find U via using matrix AA^T

Similar to how we found V. But doing it way we laid out is usually easier. We mostly found U from V rather

than doing essentially a completely different calculation for U. And indeed it would be a totally different calculation as AA^T is not at all the same as A^TA, usually not even the same size.

Why do these U, V work? Well both are orthogonal of proper size so that's fine. So just a matter of whether Σ works (and what it equals!). To that end check the matrix U^TAV:

The columns of AV are: $Av_1, Av_2, \ldots, Av_n$. So then

the ij-th entry of U^TAV is $u_i \cdot Av_j$ since the

i-th row of U^T is u_i and the j-th column is Av_j

So what does this equal?

(1) For $j > r$, since $\|Av_j\| = \sigma_j$ for all j so $Av_j = 0$
thus $u_i \cdot Av_j = 0$

(2) For $1 \leq j \leq r$, $i > r$ we have

$$u_i \cdot Av_j = u_i \cdot \sigma_j u_j = \sigma_j (u_i \cdot u_j)$$

since the u_i's are an orthonormal set, and $i \neq j$
given the range of each here.

(3) For $1 \leq i, j \leq r$ similarly to (2):

$$u_i \cdot Av_j = \sigma_j (u_i \cdot u_j) = \begin{cases} \sigma_j & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Hence overall the ij-th entry of $U^T AV$ is

$$\begin{cases} \sigma_j & \text{if } i = j \text{ and both } \leq r \\ 0 & \text{otherwise} \end{cases}$$

Thus U^TAV is a diagonal matrix (m x n), call it Σ

Then

$$U^TAV = \Sigma$$

---->

$$UU^TAV = U\Sigma$$

---->

$$AV = U\Sigma$$ since U orthogonal

---->

$$AVV^T = U\Sigma V^T$$

---->

$$A = U\Sigma V^T$$ since V orthogonal

SVD

So we have created a SVD as all three matrices in this breakdown as desired. Let's sum up what's involved in creating SVD from scratch.

(1) Compute $A^T A$

(2) Find eigenvalues of $A^T A$ which are always real and nonnegative, denoted in decreasing order as $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, each $\sigma_i \geq 0$

(3) Find an orthonormal basis of eigenvectors from this, denoted $\{v_1, v_2, \ldots, v_n\}$ with each v_i corresponding to σ_i^2. Use Gram-Schmidt if needed to create this. If some eigenvalue is repeated, that might be needed within that eigenspace.

(4) Compute $u_i = \frac{1}{\sigma_i} A v_i$ for each $\sigma_i \neq 0$

(5) The u_i's are an orthonormal set in $\mathbb{R}^m$. If not a basis for the full space, extend them to be such, i.e. $\{u_1, u_2, \ldots, u_r, u_{r+1}, \ldots, u_m\}$

Note that depending on how you extend this set to a basis for $\mathbb{R}^m$, it might require Gram-Schmidt again to make it orthogonal. Say you take the current vectors and make them the rows of a matrix then find a basis for the null space of that matrix, that basis may not be orthogonal, you will then have to make it such.

(6) Then

$$\left.\begin{aligned} U &= [u_1\ u_2\ \ldots\ u_m] \\ V &= [v_1\ v_2\ \ldots\ v_n] \end{aligned}\right\} \text{columns}$$

and Σ is the $m \times n$ "diagonal" matrix with diagonal entries: $\sigma_1, \sigma_2, \ldots, \sigma_n$

Note that if $m < n$ so that there are only m diagonal entries in A, then only will need to use $\sigma_1, \sigma_2, \ldots, \sigma_m$

In that situation that $A^T A$ is $n \times n$ with maximum rank m so beyond the first m, all those extra eigenvalues of $A^T A$ must be 0 anyway so not like you are not utilizing some more pertinent singular values,

you are just throwing out some 0s.

In diagonalization the entries on the diagonal of D are the eigenvalues of A while in SVD the entries on the diagonal of Σ are the singular values of A (i.e. the square roots of the eigenvalues of $A^T A$).

Hence another reason to use singular values rather eigenvalues as notationally that meshes better with diagonalization.

So that basically sums up the full process of finding a SVD of an aribtrary matrix.

Only thing left is to go through an example fully. Very involved but have to be ready to do that! Will do that next time.